

Final Project Deployment & Presentation Assignment

Course: Full Stack Web Development

Objective: Deploy your complete application and prepare it for final evaluation.

Overview

In this final stage, you will deploy the full-stack application developed throughout the semester. Your system must be publicly accessible and fully operational. The frontend, backend, database, WebSocket functionality, and AI component must all be connected and working together.

Deployment Requirements

1. Frontend Deployment

Compile the React frontend into a production build (dist/build folder as demonstrated in class). Deploy the frontend to Render and verify that it is publicly accessible.

2. Backend Deployment

Deploy the Node.js + Express backend to Render. Configure all required environment variables and ensure that all API endpoints are operational.

3. Database Deployment

Deploy the MySQL database to AWS RDS. The database must be reachable from external networks and must remain available during the final presentations.

4. System Integration

Configure and connect all components exactly as demonstrated during the course:

- Frontend ↔ Backend
- Backend ↔ MySQL Database
- Frontend ↔ WebSocket Service
- Frontend/Backend ↔ AI Functionality

All deployed services must function together as one integrated application.

Final System Requirements

By the end of the project, your application must include:

- React frontend
- Node.js + Express backend
- MySQL database
- ORM integration

- WebSocket functionality
- AI-powered feature
- Authentication and login flow
- Fully connected deployment environment

The application must not contain obvious UI bugs, broken functionality, navigation issues, or logical errors during the presentation.

Presentation Requirements

Each team will present its project in a short live demonstration.

Duration	Activity
20 seconds	Explain the project idea and purpose.
3 minutes	Live demonstration of the application workflow.
40 seconds	Demonstrate the database using MySQL Workbench.
45 seconds	Questions from the teaching staff.
Total: 4:45 minutes	

Idea explanation & demonstration start

To start the demonstration, one team member will explain the project idea and purpose, while another member will connect to the website using the Render URL.

Required Live Demonstration Workflow (3 Minutes)

During the demonstration, teams must:

1. Create a new user.
2. Log in.
3. Navigate to the main application page.
4. Use a primary application feature.
5. Use the AI-powered functionality.
6. Use the WebSocket functionality.
7. Navigate to the settings page.
8. Modify at least one setting.
9. Use the navigation bar to navigate to a page.
10. Log out.

Database Demonstration (40 Seconds)

Teams must connect to their AWS RDS database using MySQL Workbench and demonstrate that application usage has created or modified records in the database (new user, new message/resource, etc.)

External Access Requirements

Presentations will be conducted using a separate computer provided by the teaching staff.

This computer will have MySQL Workbench 8.0 CE installed.

The following must therefore be accessible from outside your own network:

- Render-hosted frontend
- Render-hosted backend
- AWS RDS MySQL database

Teams are responsible for testing this configuration before presentation day.

Submission Requirements

In the final project submission form (שיבוץ להצגת פרויקט), teams must provide:

- Public website URL
- Backend URL (if different)
- AWS RDS endpoint
- Database username
- Database password

Failure to provide valid credentials may compromise the evaluation of your project.

Evaluation Readiness Checklist

Before presenting, verify that:

- ☐ Website loads correctly from Render URL.
- ☐ User registration works.
- ☐ Login works.
- ☐ Main functionality works.
- ☐ AI feature works.
- ☐ WebSocket feature works.
- ☐ Settings page works.
- ☐ Database updates correctly.
- ☐ Application is accessible from an external computer.
- ☐ Database is accessible from MySQL Workbench 8.0 CE on external computer.
- ☐ Team members know how to disconnect from existing Workbench connection and create a new connection
- ☐ **All actions are timed and match the time requirements for the presentation**